

Impacts of non-native plant and animal invaders on gap regeneration in a protected boreal forest

Jessica M. Humber · Luise Hermanutz

Received: 31 August 2010 / Accepted: 25 June 2011 / Published online: 9 August 2011
© Springer Science+Business Media B.V. 2011

Abstract In balsam fir (*Abies balsamea*)-dominated boreal forests of Gros Morne National Park, Newfoundland (Canada), non-native *Cirsium arvense* (Canada thistle) has invaded forest gaps. Its management is complicated by the lack of viable control techniques and an overarching issue of gap regeneration failure attributed to browsing by non-native moose (*Alces alces*). This study identifies the impacts of thistle invasion on balsam fir regeneration and explores protocols to re-establish fir in gaps invaded by thistle and moose. Fir seeds were planted into ten gaps (five natural; five anthropogenic) and the emergence, growth, herbivory damage, and survival of fir was determined for 2 years amongst five treatments ($n = 50$ plots; 32 seeds/plot): (1) thistle monocultures in gaps; (2) where aboveground thistle biomass was removed; (3) where above- and

below-ground thistle biomass was removed; (4) non-invaded areas in gaps; and, (5) adjacent uninvaded forest edges. In addition, 432 fir seedlings (aged 15 months) were transplanted into four forest gaps within the above treatments and followed for 1 year. Results indicate that invasion of *C. arvense* negatively affects fir emergence and early survival, and may further contribute to continued balsam fir regeneration failure independent of future moose densities. However, older fir seedlings transplanted into thistle monocultures experienced a positive facilitative effect due to the protection thistle provided against small mammal herbivory. Restoration actions that combine moose density reductions with the planting of fir seedlings offers the most viable long-term strategy to re-establish the native forest canopy in thistle-invaded gaps and would likely lead to the eventual decline of shade-intolerant *C. arvense*.

Electronic supplementary material The online version of this article (doi:10.1007/s10530-011-0048-1) contains supplementary material, which is available to authorized users.

J. M. Humber · L. Hermanutz
Department of Biology, Memorial University of
Newfoundland, St. John's, NL A1B 3X9, Canada

Present Address:

J. M. Humber (✉)
Wildlife Division, Department of Environment &
Conservation, Government of Newfoundland and
Labrador, 117 Riverside Drive, P.O. Box 2007, Corner
Brook, NL A2H 7S1, Canada
e-mail: jhumber15@hotmail.com

Keywords Non-native invasion · Forest
regeneration · *Alces alces* · *Cirsium arvense* ·
Protected area · Disturbance · *Abies balsamea*

Introduction

Non-native plant invasions represent a severe threat to native species diversity, ecosystem structure and function, and human welfare, both locally and on a global scale (Vitousek et al. 1997; Levine et al. 2003). With the increased rate of movement of

species throughout the world and high rates of disturbance, non-native plants are becoming a larger component of regional floras and are prompting conservation concern by managers of protected areas worldwide (MacDonald et al. 1986; Cowie and Werner 1993; Myers and Bazely 2003). Developing a better understanding of the impacts of non-native species on the ecological integrity of protected areas (Parks Canada Agency 2000) and devising methods to reverse the effects of invasive species (Hobbs and Norton 1996; Zavaleta et al. 2001; Coomes et al. 2003) are increasingly recognized as imperative goals of conservation management.

Many of Canada's National Parks contain hundreds of introduced plant species (Mosquin 1997). The large majority of these plants are found primarily in anthropogenically-disturbed areas such as roadsides, picnic areas, campgrounds, ditches, gravel pits, and hiking trails (Mosquin 1997; Rose 2002). These source populations of non-native plants may threaten ecosystem functioning and native species diversity if they become 'invasive' and successfully expand their range into adjacent native plant communities or high-quality protected ecosystems that are remote from human activity (Burke and Grime 1996; Mosquin 1997; Rose and Hermanutz 2004; Hendrickson et al. 2005). When widespread invasion of natural areas occurs, controlling or eradicating the invader while minimizing damage to non-target native species and protecting surrounding natural communities can be a complex, and often impossible task (Myers and Bazely 2003). While invasive species control is often effective at reducing densities of non-native plants, accumulating evidence suggests that implementation of invasive alien plant control efforts in natural areas can have unexpected negative impacts on non-target native species and ecosystems (NRC 2000; Zavaleta et al. 2001; Smith et al. 2006). In fact, control and eradication practices such as chemical, mechanical, or biological control can result in more severe impacts on native species diversity and ecosystem function than by the non-native invader itself (Hobbs and Humphries 1995; Matarczyk et al. 2002; Pearson and Callaway 2003; Messing and Wright 2006) and may lead to severe environmental degradation (NRC 2000) and further exotic invasions (Smith et al. 2006).

How to remove exotic plants is a problem faced by managers of Gros Morne National Park of Canada

(Newfoundland; GMNP) where *Cirsium arvense* (L.) Scop. (i.e. creeping thistle; Canada thistle), an herbaceous perennial alien plant in North America, has recently invaded widely-dispersed natural and anthropogenic boreal forest gaps on an extensive landscape scale. Surveys throughout the Park (1,805 km²) in 2005 indicated its presence in 55% of naturally-disturbed gaps and 41% of gaps formed by small-scale timber harvesting (Parks Canada, unpublished data) at densities up to 48 shoots/m² (Humber 2009). Many of these gaps are isolated within dense coniferous canopy and are located many kilometers from the nearest road. Despite its notoriety as a noxious weed in agricultural landscapes (Moore 1975; Donald 1990; Skinner et al. 2000), *C. arvense* invasion into GMNP forests was not anticipated since it is shade intolerant (Bakker 1960; Donald 1994; Nuzzo 1997; Edwards et al. 2000; Tiley 2010) and has not generally been considered a threat to boreal forests (Haber 1997). Forests act as a relatively continuous barrier to wind dispersal of alien plant seeds (Brothers and Spingarn 1992) and the low light availability, soil pH, and nutrient availability in undisturbed forest canopies, which often separate gap openings by great distances, are unsuitable for shade-intolerant herbs such as *C. arvense* (Bakker 1960; Haber 1997; Rose 2002). In GMNP, it appears that these obstacles to invasion have been overcome as a result of direct and indirect facilitation by the high density of non-native moose (*Alces alces* L.; Rose and Hermanutz 2004; Gosse et al. 2011).

Since the Park's establishment in 1973 and the onset of protection from hunting, moose populations in GMNP have increased from 0.14 moose/km² in 1971 to densities of 5.0 moose/km² in preferred lowland forests in 2007 (Burzynski et al. 2005; Thompson 2007). In the highest density survey units, densities were as high as 12.0 moose/km² in 2007 (Thompson 2007)—representing some of the world's highest densities (McLaren et al. 2000). Browsing by these high density moose populations within GMNP's forests is having deleterious impacts on native tree species diversity and is threatening forest regeneration (McLaren et al. 2004; Burzynski et al. 2005; Forbes 2006). In winter, moose preferentially browse the advanced regeneration (i.e. seedling bank) of balsam fir (*Abies balsamea* (L.) Mill.), the dominant species in GMNP's forests and in all productive upland sites throughout western

Newfoundland (Hall and Richardson 1973; Burzynski et al. 2005). The loss of the balsam fir seedling bank is highly detrimental to post-disturbance stand regeneration because it is the species' primary means of restocking gaps after disturbance (Duchesneau and Morin 1999) and balsam fir seeds do not form a persistent seed bank (Morin and Laprise 1997; Greene et al. 1999). Densities of balsam fir seedlings within gaps are insufficient to restock forests (<1 individual per m^2), and where young balsam fir does occur, it is highly stunted from sustained browsing pressure and is unable to reach adult reproductive stages or form a canopy (Humber 2009).

Moose also facilitate and promote the invasion of non-regenerating gaps by alien plants such as *C. arvense* by maintaining disturbed conditions. After canopy opening, moose maintain gap susceptibility to invasion by focusing their browsing on the highly palatable species that revegetate early successional stands (Rose and Hermanutz 2004; Forbes 2006; Gosse 2006; Gosse et al. 2011). This activity combined with trampling of native vegetation maintains high light availability and creates competition-free, bare ground sites. Moose in GMNP also act as the primary conduits for alien seed dispersal into these favorable isolated habitat patches (Rose and Hermanutz 2004) by transporting seeds on their hooves and fur along moose travel corridors.

Instead of investigating costly and potentially ineffective and harmful methods of directly controlling *C. arvense* in GMNP, this study takes a broader ecosystem-scale approach to management (Noss 1996). It is yet unclear whether the accompanying widespread invasion of *C. arvense* into open boreal forest gaps in GMNP will continue to impede the expected successional trajectory to a closed canopy forest should moose undergo an eventual population crash or should human intervention be implemented to decrease moose populations (McLaren et al. 2004; Forbes 2006; McLaren and Tom-Dery 2007). Under similar circumstances, Coomes et al. (2003) found that, when *not* combined with active replanting of native plant communities, culling programs aimed at controlling invasive deer populations in New Zealand have not resulted in recovery of the original forest to its pre-invasion composition; rather, introduced plants (e.g. exotic grasses) continue to impede reestablishment of native forest species independent of deer populations. Therefore, the primary goal of

this study is to better understand if, and how, *C. arvense* affects balsam fir regeneration processes. The second goal was to determine if direct sowing of balsam fir seeds or planting of greenhouse-established fir seedlings would be successful in actively promoting native forest regeneration within thistle-invaded gaps. The final goal of this study was to identify how factors influencing balsam fir regeneration differ among gaps formed by natural or anthropogenic processes, as this knowledge could help guide management actions.

To achieve these outlined objectives, both a balsam fir seed-addition and a seedling-addition experiment were performed in naturally and anthropogenically disturbed forest gaps within GMNP and success of this native conifer (i.e. emergence, growth, herbivory damage, and survival) was compared among regions of each gap invaded with *C. arvense* monocultures, areas not containing *C. arvense* growth, and the adjacent forest edge. To our knowledge, this is the first study to experimentally examine native forest regeneration within adjacent areas concurrently containing or lacking an alien plant infestation. As balsam fir establishment success was hypothesized to be poor within *C. arvense* monocultures relative to uninvaded gap and forest edge areas, potential mechanisms by which *C. arvense* might impede the establishment of balsam fir were also explored. Balsam fir success in unaltered *C. arvense* monocultures was compared to success in plots where only thistle shoots were removed, excluding aboveground competition for light and potential aboveground allelopathic effects described for thistle shoots and leaves (Kazinczi et al. 2001; Glinwood et al. 2004; Humber 2009) but maintaining belowground competition by roots, potential residual allelochemicals in soil, and any active allelochemical interference by the remaining roots (Kazinczi et al. 2001); as described in other plant species, biomass removal can either stimulate chemical defenses (Siemens et al. 2002) or result in a tradeoff between chemical defense and shoot regrowth (Hartley and Jones 1997). It was hypothesized that success of fir would not be improved significantly by removing *C. arvense* shoots alone because new balsam fir emergents and older seedlings are shade tolerant (Hall and Richardson 1973; Burns and Honkala 1990). Additionally, fir success was compared to plots where all competitive and potential direct allelopathic interference was removed by

eliminating above- and below-ground thistle biomass (Bendall 1975; Kazinczi et al. 2001) but retaining any residual toxicity in the soil. It was hypothesized that fir success would be improved in this treatment but would remain lower than in uninvaded areas since studies of allelopathy in *C. arvense* have demonstrated potential for residual soil to exclude neighboring vegetation (Bendall 1975; Stowe 1979; Wilson 1981).

Methods

Study area

This study was conducted in the boreal forest of Gros Morne National Park, located on the western coast of the Great Northern Peninsula on the island of Newfoundland, Canada (49°30' N, 57°50' W). Our research was restricted to lowland forests (<150 m elevation) which comprises 43% of the park's total area (1,805 km²; Bouchard et al. 1991; Rose and Hermanutz 2004). These forests are dominated by balsam fir, white spruce (*Picea glauca* (Moench) Voss), black spruce (*P. mariana* (Mill.) B.S.P.), and white birch (*Betula papyrifera* Marsh). Soils in the park are primarily humo-ferric podzols characterized by poorly drained acidic, mineral soils and formed under the influences of the parent material, humid climate, and coniferous vegetation (Clayton et al. 1977).

All experiments were performed within 10 disturbed canopy gaps distributed widely throughout the Park's lowland forest (Online Resource 1). Forest gaps were identified using time-sequenced GIS-indexed aerial photographs (ArcGIS version 9.1, ESRI Corporation, Redlands, California) obtained from Parks Canada as well as with the aid of local foresters. All chosen sites were originally fir-dominated but have not yet recovered after disturbance. Gaps chosen were characteristic of the region; 44.5% of the disturbed lowland forest in GMNP (95.2 km²) contains 'severely impaired/failed' forest regeneration typical of the forest gaps chosen in this study, while another 24.1% of disturbed forest is considered 'vulnerable' to regeneration failure due to poor canopy development and varying success of understory regeneration (Parks Canada, unpublished data). Gaps were chosen based on their predominant disturbance regime: half of the gaps ($n = 5$) were

formed naturally by insect defoliators including eastern spruce budworm (*Choristoneura fumiferana* Clem.) and hemlock looper (*Lambdina fiscellaria fiscellaria* Guenee) and half were formed by legal small-scale, traditional domestic harvesting by local residents (i.e. anthropogenic disturbance). As part of the Federal-Provincial Agreement for GMNP, local residents within the Park's seven enclave communities and two bordering communities are permitted to cut wood for personal uses within 12 Domestic Timber Harvest blocks comprising 129 km² of the 308 km² of lowland forest (Burzynski et al. 2005). Time since disturbance varied among gaps (ranging from 6 to 29 years; Online Resource 1), with all gaps having been formed within the past 30 years. Timber harvesting, or tree fall after succumbing to insect defoliation, often occurred over multiple years after the initial year of disturbance to eventually result in complete canopy removal. In addition to the predominant disturbance regime and time since disturbance, site selection was based upon the presence of *C. arvense* monocultures in gaps, accessibility of gaps (i.e. normally < 1.5 km of a road, with the exception of sites SPA and SPB which were reached by boat; Online Resource 1), approximate size of the gaps (i.e. between 1 and 4 hectares), and to achieve a wide distribution of gaps across the park's lowland forest area.

The undisturbed forest canopies surrounding the gaps were dominated by balsam fir mixed with subdominant white spruce and white birch; less abundant species included trembling aspen (*Populus tremuloides* Michx.), mountain maple (*Acer spicatum* Lam.), and black spruce. The ground vegetation in forests included a wide variety of herbs and mosses including *Pleurozium schreberi* (Brid.) Mitt., *Ptilium crista-castrensis* (Hedw.) De Not, *Hylocomium splendens* (Hedw.) B.S.G., *Rhytidiadelphus* sp., and *Dicranum* spp.

Seed addition experiment

In the first week of July 2006, five experimental plots (0.5 m × 0.5 m) were established within each forest gap ($n = 50$ total plots) for addition of balsam fir seeds. Each plot was located within a specific region of the gap (noted below) and was considered a specific 'treatment' in which success of seed addition could be determined:

- (i) *C. arvense* monoculture (T): Dense, unmanipulated *C. arvense* monocultures with an average density of 30 shoots/m². At the time of planting, thistle shoot height averaged 0.9 m among sites, although heights frequently reached >1.5 m later in the summer.
- (ii) *C. arvense* monoculture with aboveground biomass removed (T_{-A}): Thistle monoculture as in i), but from which all aboveground *C. arvense* biomass was removed by cutting shoots at soil level immediately prior to planting of balsam fir seeds and continually throughout the growing season at 10 day intervals.
- (iii) *C. arvense* monoculture with above- and below-ground biomass removed (T_{-AB}): Both aboveground and belowground *C. arvense* biomass were completely removed by digging up individual plants and their root systems from the soil immediately prior to planting balsam fir seeds; only aboveground thistle regrowth was subsequently cut at 10 day intervals to prevent damage to sensitive roots of balsam fir seedlings.
- (iv) Control—Uninvaded region of the gap (UG): A region within the gap that did not contain *C. arvense* and was located at least 15 m from the nearest thistle plants. To minimize variability among gap replicates, this treatment was always established within a seedbed of *Cornus canadensis* L., a native plant that is ubiquitous throughout the boreal forest, regularly co-occurs with balsam fir, and is known as a favorable seedbed for balsam fir establishment (Hall and Richardson 1973; Burns and Honkala 1990).
- (v) Control—Uninvaded forest edge (UE): A plot randomly established 5 m into the adjacent, uninvaded forest edge; this treatment was established as a control for the gap itself, with which success of balsam fir seed addition in the UG treatment was compared.

Within a *C. arvense* monoculture, the three plot locations for the T, T_A , and T_{-AB} treatments (i–iii above) were randomly established within the same patch at a minimum separation of 1 m and treatment was assigned randomly to each plot.

Due to low local balsam fir seed crop in the year prior to study, all seeds were obtained from the

National Tree Seed Centre (NTSC), Canadian Forest Service (Fredericton, NB). Seeds were taken from trees in two Newfoundland populations (Bay D’Espoir and Roddickton) and had documented germination rates of 50.0%. As per NTSC protocol, seeds were moist-stratified at 4°C for 3 weeks prior to planting; the documented germination rates were confirmed through controlled germination tests of seeds in environmental chambers. As the target was to establish and monitor up to 16 fir seedlings per plot, 32 balsam fir seeds (i.e. with 50% germination rates) were planted into each 0.5 m × 0.5 m plot by gently pushing them into the exposed mineral soil surface ($n = 1,600$ total seeds) immediately after plot preparation. Seeds were planted in pairs in a 4 × 4 grid, with 10 cm spacing between each group of two seeds and a 10 cm unplanted border on all sides. Immediately adjacent to each planted plot, an equally-sized (0.5 m × 0.5 m) unseeded control plot prepared in the same manner was established to quantify natural balsam fir emergence and establishment.

Throughout the first summer (2006), plots were surveyed for newly emerged seedlings at 10 day intervals. Overall emergence rates were based on detection of the shoot above the soil surface. The timing of seedling emergence was observed to closely mimic the period of natural balsam fir emergence in the region. Newly-emerged seedlings were thinned to no more than one seedling per 10 cm × 10 cm area by clipping at the base of the stem, for a maximum remaining density of 16 seedlings per plot. At 10 day intervals, seedlings were monitored for survival and occurrence of browse damage until the end of the field season on 29 August 2006, at which time the season’s growth had ceased and the terminal bud had set. At this time, height of remaining seedlings was determined to the nearest 0.001 cm. All seedlings were re-monitored during mid-June of 2007 and late May of 2008, or approximately 1 and 2 years after planting, to track survival.

Seedling addition experiment

In mid-June of 2007, 432 balsam fir seedlings aged 15 months old were planted into four forest gaps within GMNP. Seedlings were obtained from Wooddale Provincial Tree Nursery, Newfoundland (Department of Natural Resources, Government of NL) and

originated from balsam fir seeds collected from western Newfoundland. At planting, seedlings averaged 24.7 ± 0.2 cm (SE) in height. Gaps chosen for planting of seedlings were a subset of those having received seed addition. Two of the sites were naturally disturbed by insect outbreaks (i.e. sites BBA and SPB; Online Resource 1) and two were harvested sites (i.e. sites MBB and TKB). Each gap contained multiple *C. arvense* patches, together comprising approximately one third of the overall gap area. In each site, seedlings were planted into one of four locations corresponding to four of the seed-addition treatments (i, ii, iv, and v above), with the exception of the T_{AB} treatment due to limited seedling availability. Three replicate plots were established for each treatment and randomly positioned at a minimum separation of 1 m. Each plot measured 4 m $\times$ 4 m, with a 1 m unplanted border on all four edges and with nine seedlings planted in a 3 $\times$ 3 grid at a distance of 1 m apart on all sides. A total of 48 plots and 432 seedlings were established among the four sites. Each seedling was identified using a small metal tag pinned to the ground near the base of the stem. Change in seedling height (to 0.1 cm), change in basal diameter (to 0.001 cm), presence of browsing or desiccation damage, and mortality were monitored near the end of the first summer growing season (18–20 August 2007) and again in late May of 2008 after almost 1 year in the field to determine changes over the autumn–winter period.

Abiotic soil properties

Soil samples were collected for soil moisture determination from each of the 10 gaps on: 14–16 July 2006, 3–5 August 2006, and 23–25 August 2006; and from eight of the gaps on 13–15 June 2007 (two of the anthropogenic gaps could not be sampled at this time due to the presence of black bears). In each gap, a 1 m² quadrat was randomly established in three treatments: *C. arvense* monoculture (T), uninvaded gap (UG), and uninvaded edge (UE). Soil cores to 10 cm depth were collected from the four corners of the quadrat and pooled in a tightly sealed sample bag at 4°C until laboratory procedures to determine percent soil moisture could be performed. At this time, pooled soil samples from each quadrat were thoroughly mixed and three subsamples of 10 ± 0.1 g were removed, weighed, and dried in a

70°C drying oven for 48 h before reweighing. Percent soil moisture was calculated as:

$$\% \text{ soil moisture} = (\text{wet mass} - \text{dry mass}) / (\text{dry mass}) \times 100$$

Soil temperature at 10 cm depth was also determined in the eight gaps on 13–15 June 2007 using a Barnant (Type K) Thermocouple Thermometer (Barnant Company, Barrington, IL). As in the soil moisture procedure, in each gap a 1 m² quadrat was randomly established within three treatments (i.e. T, UG, and UE) and soil temperature (°C) at 10 cm depth was measured at each of the four corners of each quadrat. Soil moisture and temperature data were used to compare environmental conditions among treatments within each gap.

Statistical analyses

All statistical analyses were performed on SAS version 9.1 (SAS Institute, Cary, NC, USA) using generalized linear mixed modeling (GLMM; McCullagh and Nelder 1999). Where the response variable was a continuous variable with normal errors (i.e. height, basal diameter, soil moisture, or soil temperature), a mixed-model analysis of variance was performed using the MIXED procedure; random effects are specified in each model as outlined below. All test assumptions were met as determined through examination of residuals for normality, homogeneity, and independence. In general, where the response variable consisted of binary or proportional data (i.e. proportion of seedlings emerging, surviving, sustaining browse damage, or desiccating) with non-normal errors, mixed-model logistic regression was implemented in the GLIMMIX procedure with a logit link and binomial errors. The MIXED and GLIMMIX procedures have a comprehensive set of tools for working with random effects (Littell et al. 2002). Variance components were estimated by restricted maximum likelihood (REML) in the MIXED procedure and residual pseudo-likelihood in the GLIMMIX procedure. Denominator degrees of freedom were calculated using the Satterthwaite approximation described in Fai and Cornelius (1996) except where repeated measures were incorporated into the model design, in which case the Kenward-Rogers approximation was used instead to avoid inflated Type I

errors (Littell et al. 2002). Analyses were considered significant at $\alpha = 0.05$.

Seed addition and seedling addition experiments

The main model for the seed and seedling addition analyses consisted of both fixed and random categorical variables. Fixed factors included treatment (T, T_{-A}, T_{-AB}, UG, or UE; excluding T_{-AB} in the seedling addition), disturbance regime (natural or anthropogenic), and their interaction. The variable ‘gap’ was nested within disturbance regime and treated as a random variable to account for variation among individual gaps. To handle the non-independence of seed/new emergent clustering within plots in the seed addition experiment (i.e. spatially repeated measures), ‘plot identity’ was included as a repeated effect (equivalently, a ‘residual’ or ‘R-side’ random effect in GLIMMIX) and a compound symmetric (i.e. exchangeable) relationship was specified among the individuals within a plot to account for correlated errors (Horton and Lipsitz 1999). In the seedling addition experiment, each seedling within a plot was considered independent as it was widely separated from other individuals by at least 1 m on all sides, however, a second random variable accounted for treatment replication within any gap ($n = 3$ plots per each treatment in each gap) and was nested within gap.

In the seed addition experiment, overall balsam fir emergence was analyzed based on the proportion of viable seeds planted (i.e. 32 seeds/plot $\times$ 50% viability = 16 viable seeds/plot). Cumulative fir seedling survival was determined at three time periods: at the end of the first growing season (i.e. ‘GS1’), at approximately 1 year (i.e. ‘Y1’), and after about 2 years (i.e. ‘Y2’). Seedling survival at each of these periods was expressed as the proportion of the total number of seedlings initially present in the plot (i.e. those remaining after post-emergence thinning). Fir survival was also determined between the first autumn–winter period (i.e. ‘AW1’, the period between GS1 and Y1) and between the first and second years (i.e. between Y1 and Y2). Cumulative browsing damage on seedlings (including lethal and non-lethal damage) and analyses of seedling height were determined at the GS1 and Y1 stages. In the seedling addition experiment, cumulative fir seedling survival was analyzed at GS1 and Y1, with autumn–winter survivorship between the GS1 and Y1 periods

also determined. Desiccation and browsing damage (lethal and non-lethal combined) as well as change in height and basal diameter since planting were analyzed at the GS1 and Y1 stages.

In all analyses for both experiments, when the effect of treatment was significant a priori linear contrasts were used to compare means among selected treatments (Online Resource 2) and Bonferroni corrected for the number of comparisons performed (seed addition: $\alpha' = 0.0125$; seedling addition: $\alpha' = 0.0167$).

Abiotic soil properties

Fixed factors in the analyses of soil moisture and temperature using the Proc MIXED procedure consisted of treatment (T, UG, or UE), disturbance regime, and their interaction. The random variable ‘gap’ was nested within disturbance regime. In the analysis of soil moisture in 2006, potentially correlated errors resulting from repeated measures over the field season were accounted for by adding ‘sampling period’ as a repeated effect with a first-order autoregressive structure to account for equal periods between re-sampling (Littell et al. 2002). Soil moisture was sampled once in 2007 and was analyzed separately. When the effect of treatment was significant, three linear contrasts were used to compare among each pair of treatment means and were Bonferroni corrected for three comparisons ($\alpha' = 0.0167$).

In all statistical analyses performed in this study, where the interaction between treatment and disturbance regime was significant, the main effects could not be examined within this model (Sokal and Rohlf 1995) and thus the effect of ‘treatment’ was examined separately within each disturbance regime. We report only significant interaction terms. Results are expressed as mean $\pm$ standard error of the mean.

Results

Seed addition experiment

Emergence

Natural balsam fir emergence within all adjacent unseeded control plots was zero, thus emergence

results reflect the outcome of seed addition, with no contribution from the resident seed bank. The effect of treatment on balsam fir emergence was determined separately within each disturbance regime due to a significant interaction ($F_{4,31} = 4.56$, $P = 0.0051$). In both anthropogenic and natural gaps, treatment had a significant effect on balsam fir emergence (harvested gaps: $F_{4,16} = 5.93$, $P = 0.0040$; natural gaps: $F_{4,15} = 5.40$, $P = 0.0067$). In the forest edges of harvested gaps, emergence of viable seeds averaged $27.5 \pm 4.99\%$ (SE), while within the gap openings the number of viable seeds emerging was extremely low, with no treatment in the gap having $\geq 10\%$ emergence regardless of whether *C. arvense* was present (Fig. 1). Total+ing all harvested gaps there was a total of only 2/80, 1/80, 8/80, and 2/80 viable seeds emerging in the T, T_{-A}, T_{-AB}, and UG treatments, respectively. Removing aboveground thistle biomass (T_{-A}) or both above- and belowground thistle biomass (T_{-AB}) did not significantly improve emergence within thistle monocultures ($P = 0.6332$ and $P = 0.1364$, respectively; Fig. 1). However, emergence in thistle monocultures was no lower than in areas of the gap not invaded by *C. arvense* ($P = 1.0000$). Only in the forest edge was emergence significantly improved over that in the gap (UE vs. UG; $P = 0.0066$; Fig. 1).

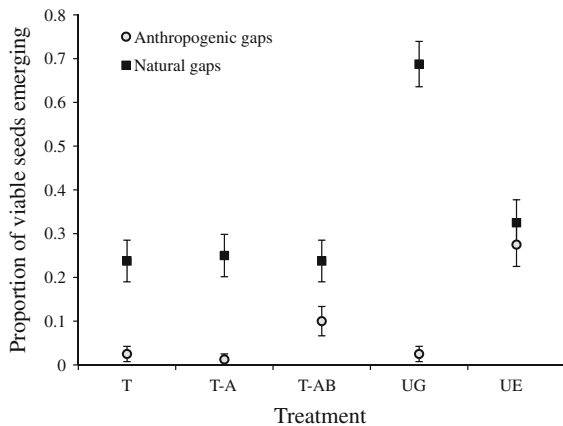


Fig. 1 Mean ($\pm$ SE) proportion of viable balsam fir seeds emerging after planting seeds during July 2006 into five field treatments in boreal forest gaps within Gros Morne National Park, NL, Canada, disturbed by either anthropogenic or natural processes. *Treatment abbreviations:* T = *Cirsium arvense* (i.e. thistle) monoculture, T_{-A} = thistle monoculture with above-ground shoot biomass removed, T_{-AB} = thistle monoculture with above- and below-ground biomass removed, UG = uninvaded region of the gap, and UE = uninvaded forest edge

Balsam fir emergence was much higher in naturally-disturbed gaps than in harvested gaps (Fig. 1). This difference was most notable in UG, where the mean emergence in naturally-disturbed gaps was 27.5 times higher than in anthropogenic gaps. Fir emergence within thistle monocultures (T) in natural gaps, although higher than the equivalent treatment in anthropogenic gaps, was only one third of the emergence in uninvaded areas (UG; $P = 0.0076$). This high emergence in UG was significantly greater than in the UE ($P = 0.0084$) treatment. Again, in natural gaps the fir emergence within *C. arvense* monocultures was not significantly increased by removal of thistle biomass (T_{-A}: $P = 0.1113$; T_{-AB}: $P = 0.1214$); however, all emergence of balsam fir seedlings in thistle treatments (T, T_{-A}, T_{-AB}) occurred within just two of the five naturally-disturbed gaps (SPA and SPB; Online Resource 1). No emergence occurred in T, T_{-A}, and T_{-AB} treatments in the other natural gaps.

Survival, browse damage, and height growth

The impacts of *C. arvense* monocultures on fir seedling survival, height, and browse damage could not be determined within anthropogenic gaps due to extremely low emergence. In naturally disturbed gaps, survival of newly emerged seedlings over the first summer growing season ranged from 32.6 to 72.2% among treatments (Fig. 2). The lowest survival was within the uninvaded gap (UG) treatment whereas the highest survival rates were in thistle monocultures (T) and were 2.25 times higher than in UG. In T_{-A} and T_{-AB} treatments, mean fir survival rates were also 1.62 and 1.80 times higher than in UG, respectively. By the following spring however (Y1), this pattern of higher survival within thistle areas was no longer evident (Fig. 2). After 2 years in the field (Y2), no seedlings remained in any thistle treatment while 21.7 and 33.3% of seedlings were still alive in UG and UE. However, none of these survival rates differ statistically.

Seedling herbivory was attributed primarily to voles because the stem was clipped cleanly; the remaining herbivory was attributed to slugs which leave characteristic slime trails. Herbivory was the main cause of mortality in the UG, UE, and T_{-AB} treatments; as no trace could be found of 59 and 44% of the seedlings within the T and T_{-A} treatments,

respectively, the cause of death was largely undetermined in these treatments (Fig. 3). Desiccation was a minor cause of seedling mortality and occurred primarily within the T_{AB} treatment.

As in the analyses of seedling survival, there were no significant differences in height (GS1: $F_{4,4} = 2.13$, $P = 0.2570$; Y1: $F_{4,16} = 1.82$, $P = 0.1723$) or browse damage (lethal and non-lethal combined; GS1: $F_{4,5} = 0.43$, $P = 0.7854$; Y1: $F_{4,4} = 0.74$, $P = 0.6069$) among treatments in naturally disturbed gaps. Yet, examination of browsing patterns among treatments reveal that herbivory pressure was lower within T plots during the GS1 period than in all other treatments (Fig. 4). Although browse damage in the three thistle treatments (T, T_A, and T_{AB}) increased during the first autumn–winter period (over which time *C. arvensis* shoots die back) while browse levels in UG and UE did not increase, by Y1 levels of browse damage still remained lower in T than in any other treatment within the gap opening (Fig. 4).

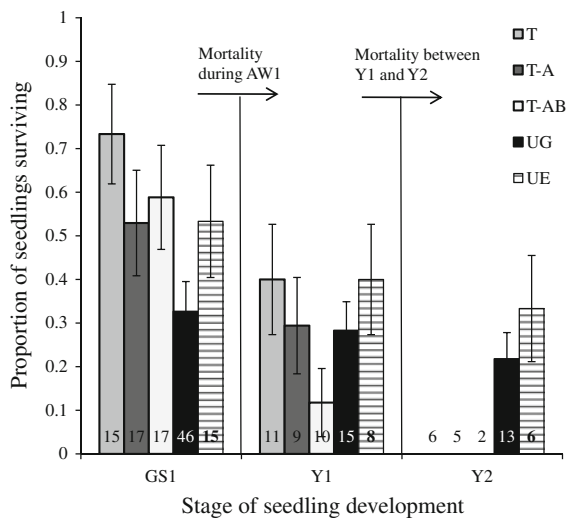


Fig. 2 Mean ($\pm$ SE) proportion of newly emerged balsam fir seedlings surviving (after seeds were sown in July 2006) in five field treatments in naturally disturbed boreal forest gaps of Gros Morne National Park, NL, Canada, at the end of the first 2006 summer growing season (GS1), first year (June 2007; Y1), and second year (May 2008; Y2). AW1 The first autumn–winter period. Sample sizes are indicated within bars and differ due to variable emergence and survival rates among treatments. Refer to Fig. 1 for treatment abbreviations

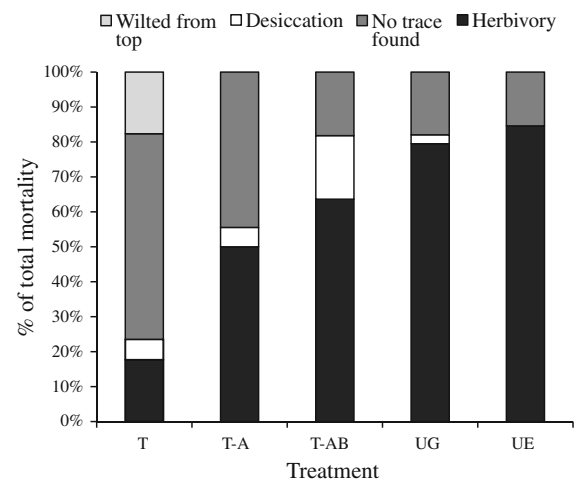


Fig. 3 Cause of mortality of balsam fir seedlings over the first 2 years of life after initial emergence in July 2006 in five field treatments in Gros Morne National Park, NL, Canada. Refer to Fig. 1 for treatment abbreviations

Seedling addition experiment

Survival

After the GS1, AW1, and Y1 periods, there was no significant difference among treatments in the survival of transplanted balsam fir seedlings (Table 1).

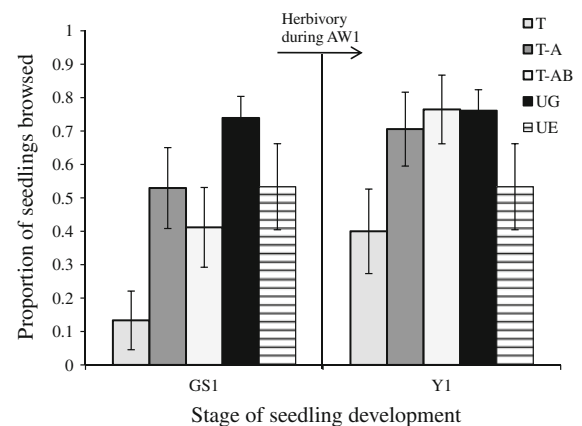


Fig. 4 Mean ($\pm$ SE) proportion of newly emerged balsam fir seedlings having browse damage in five field treatments in naturally disturbed boreal forest gaps of Gros Morne National Park, NL, Canada, at the end of the first summer growing season (August 2006; GS1) and end of the first year (June 2007; Y1) after sowing seeds in July of 2006. AW1 The first autumn–winter period. Refer to Fig. 1 for treatment abbreviations

Table 1 Mean proportion of balsam fir seedlings surviving after various periods of time following transplant into four field treatments in boreal forest gaps of Gros Morne National Park, NL, Canada

Period of seedling survival	Mean (SE) by treatment ^a				Mixed-model logistic regression		
	T	T _{-A}	UG	UE	F	df	P
GS1: Cumulative survival over first growing season	0.991 (0.009)	0.991 (0.009)	0.963 (0.018)	0.991 (0.009)	–	–	–
AW1: Autumn–winter survivorship	0.804 (0.038)	0.785 (0.040)	0.740 (0.043)	0.804 (0.038)	0.79	3, 40	0.5055
Y1: Cumulative first year survivorship	0.796 (0.039)	0.778 (0.040)	0.713 (0.044)	0.796 (0.039)	1.12	3, 40	0.3533

Results of mixed-model logistic regressions and their statistical significance among treatment means are shown. Due to the very low mortality over GS1, inferential statistics could not be performed

^a Treatment abbreviations: T = *Cirsium arvense* (i.e. thistle) monoculture, T_{-A} = thistle monoculture with aboveground shoot biomass removed, UG = uninvaded region of the gap, and UE = uninvaded forest edge

Relative to non-thistle treatments, survival of fir seedlings was never lower within *C. arvense* monocultures. Survival of balsam fir seedlings was very high over the first summer growing season; overall, only 7 of the 432 total seedlings (1.6%) experienced mortality. All but one of these recorded instances of mortality occurred within a single site (SPB) and all were a result of desiccation. Survival was also high over the AW1 period and by the following May, after approximately 1 year in the field (i.e. by Y1), overall cumulative survival was 77.1% (i.e. 333/432 seedlings).

There were no significant differences in the survival of fir seedlings at Y1 between anthropogenic and natural gaps ($F_{1,2} = 0.46$, $P = 0.5662$). In all but one site, seedling survival was greater than 80% and almost all mortality was a result of either direct herbivory or uprooting by moose (Fig. 5). There was a highly elevated mortality level (by 2.6–4.5 times) in one of the naturally-disturbed gaps (i.e. SPB) where atypically low soil moisture levels (Fig. 5) apparently resulted in excessive seedling loss to desiccation (see following section).

Desiccation

Desiccation was not a significant mortality factor for seedlings. Only 27 of the 432 seedlings (6.3%) experienced desiccation effects by the end of their first summer in the field and all but one of these occurred within the insect-disturbed gaps, which appears to be related to lower soil moisture in these areas (Fig. 5; Humber 2009). The proportion of seedlings desiccating within insect-disturbed gaps over GS1 did not significantly differ among the four

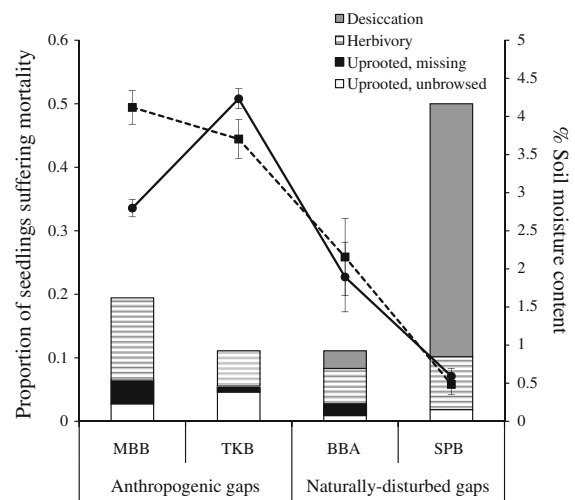


Fig. 5 Total proportion of balsam fir seedlings experiencing mortality, and the cause of this mortality, over the first year (Y1) after being transplanted at 15 months of age (in June 2007) into four boreal forest gaps in Gros Morne National Park, NL, Canada, disturbed either through anthropogenic or natural processes. Mean ($\pm$ SE) percent soil moisture content for each gap as determined in the summer growing seasons in 2006 (the previous year; dashed line with square marker) and 2007 (year of planting; solid line with circle marker) are also indicated

treatments ($F_{3,20} = 2.38$, $P = 0.1004$), although the T treatment plots had lower desiccation compared with the UE and UG plots during this period. By the end of the first year in the field (Y1), cumulative desiccation was $\sim 18\%$ (78/432 seedlings). Again, desiccation was a factor only in insect-disturbed gaps ($n = 77/78$ of the cases), with almost all instances of desiccation occurring within the SPB gap ($n = 68$). In these natural insect-disturbed gaps, desiccation

differed significantly among treatments ($F_{3,20} = 3.55$, $P = 0.0331$), with most occurring within the open uninvaded areas of the gaps (UG: $48.1 \pm 6.8\%$) where desiccation was significantly higher than in the forest edge (UE: $25.9 \pm 5.9\%$; $P = 0.0073$). Removing *C. arvensis* shoot biomass did not significantly affect the level of fir seedling desiccation (T: $29.6 \pm 6.2\%$; T_{-A}: $38.9 \pm 6.6\%$; $P = 0.2075$).

Herbivory

Herbivory on transplanted balsam fir seedlings was attributed entirely to moose browsing and was most significant over the fall and winter period. Browse damage on seedlings by moose was low (25/432) during the first summer. Although limited, the responses indicate that browsing was 3.3–5.0 times higher in areas within the gap and forest edge lacking *C. arvensis* (UG: $9.3 \pm 2.8\%$; UE: $9.3 \pm 2.8\%$) than in the T ($2.8 \pm 1.6\%$) and T_{-A} ($1.9 \pm 1.3\%$) treatments, respectively. Moose herbivory on seedlings greatly increased over the fall and winter, and by late May (Y1), 49.5% (214/432) of seedlings had been browsed by moose. Browse damage was disproportionate among treatments at Y1 ($F_{3,40} = 8.96$, $P = 0.0001$), with seedlings in the forest edge having 1.5 times more browse damage than those in uninvaded regions of the gap (UE: $70.4 \pm 4.4\%$; UG: $47.2 \pm 4.8\%$; $P = 0.0009$). Levels of herbivory in *C. arvensis* monocultures (T: $38.9 \pm 4.7\%$) did not differ from those in UG ($P = 0.2069$), and removing *C. arvensis* shoots did not significantly influence herbivory damage (T_{-A}: $41.7 \pm 4.7\%$; $P = 0.6650$). There was no difference in herbivory between

anthropogenically and naturally disturbed gaps (dist: $F_{1,2} = 0.16$, $P = 0.7266$).

Growth

All analyses of balsam fir seedling growth were performed after exclusion of all browsed seedlings to avoid confounding effects of herbivory on height with any potential treatment effect. The effect of treatment on fir height growth after the first summer growing season (GS1) was determined separately within each disturbance regime due to a significant interaction term (Table 2). In anthropogenic gaps, change in seedling height after planting differed significantly amongst treatments ($P < 0.0001$; Table 3), with seedlings in the T treatment gaining significantly more height than those T_{-A} plots and marginally more height than those in UG plots. There was no difference in seedling height growth between UG and UE treatments (Table 3). In natural gaps, the differences in seedling height growth amongst treatments was significant ($P = 0.0144$, Table 3) and seedlings in *C. arvensis* monocultures experienced twice the height growth of seedlings in UG and UE treatments (2.01 and 2.02 times more, respectively; Table 3). Removing *C. arvensis* shoots did not affect seedling heights as it did in anthropogenic gaps. In all treatments except T_{-A}, the height growth of seedlings over GS1 in the naturally-formed gaps was less than half the increase in height in the harvested gaps (Table 3). There were no significant effects of treatment or disturbance regime on (unbrowsed) balsam fir basal diameter growth in any time period, with one exception in natural gaps (GS1 period).

Table 2 Statistical results of mixed-model ANOVAs showing the fixed effects of treatment, disturbance regime, and their interaction on change in balsam fir seedling height during the

first growing season (GS1) and first year (Y1) after transplant into boreal forest gaps in Gros Morne National Park, NL, Canada

Time period	Variable	df	F	P
GS1	Treatment	3, 382	4.19	0.0062*
	Dist. regime	1, 2	3.55	0.2001
	Treatment*dist. regime	3, 382	7.72	<0.0001*
Y1	Treatment	3, 185	1.74	0.1603
	Dist. regime	1, 2	2.38	0.2681
	Treatment*dist. regime	3, 185	1.16	0.3254

Seedlings damaged by browsing were excluded from statistical analysis

* Results significant at $\alpha < 0.05$

Table 3 Statistical results of mixed-model ANOVAs comparing the effect of treatment in which balsam fir seedlings were transplanted at 15 months of age on the mean (SE) change in

seedling height over the first summer growing season (GS1) within natural and anthropogenic boreal forest gaps in Gros Morne National Park, NL, Canada

	Mean (SE) change in seedling height (cm) over GS1				Difference among treatments ^a			A priori contrasts ^a			
	T	T _{-A}	UG	UE	df	F	P	T – T _{-A}	T – UG	UG – UE	T – UE
<i>Disturbance</i>											
Anthropogenic	3.09 (0.18)	1.62 (0.20)	2.39 (0.26)	2.72 (0.28)	3, 198	8.16	<0.0001	<0.0001*	0.0313	0.3618	0.2245
Natural	1.54 (0.28)	1.61 (0.21)	0.77 (0.32)	0.76 (0.19)	3, 187	3.61	0.0144	0.7289	0.0865	0.5408	0.0163*

Seedlings damaged by browsing were excluded from statistical analysis. Where the effect of treatment is significant at $\alpha < 0.05$, results of a priori contrasts are shown

* Contrast is statistically significant after Bonferroni correction

^a Treatment abbreviations: T = *Cirsium arvense* (i.e. thistle) monoculture, T_{-A} = thistle monoculture with aboveground shoot biomass removed, UG = uninvaded region of the gap, and UE = uninvaded forest edge

Abiotic parameters

In 2006, soil moisture levels at seedling rooting depth did not differ among treatments ($F_{2,42} = 0.36$, $P = 0.7023$) but there was a significant difference in percent soil moisture content among disturbance regimes ($F_{1,8} = 9.33$, $P = 0.0159$). Soil of naturally disturbed gaps was considerably drier than anthropogenic gaps (natural: $1.24 \pm 0.42\%$ moisture; anthropogenic: $3.06 \pm 0.42\%$); with one exception, soil moisture content was higher in all anthropogenic gaps than in natural gaps. In 2007, soil moisture levels in each treatment was determined separately for each disturbance regime due to a significant interaction ($F_{2,57} = 8.72$, $P = 0.0005$). In anthropogenic gaps, there were marginally significant differences among treatments ($F_{2,22} = 3.37$, $P = 0.0531$), with % soil moisture content in T ($3.26 \pm 0.49\%$) marginally higher than in UE ($2.79 \pm 0.49\%$; $P = 0.0290$) and UG ($2.83 \pm 0.49\%$; $P = 0.0432$) treatments. In natural gaps, soil moisture content among treatments was significantly different ($F_{2,36} = 14.38$, $P < 0.0001$) with the highest moisture levels in UE ($1.72 \pm 0.28\%$), less moisture in T ($1.10 \pm 0.27\%$), and very low moisture levels in UG ($0.51 \pm 0.27\%$); all treatments were significantly different at $\alpha' = 0.0167$ when Bonferroni-corrected for three comparisons. In 2007, soil in the five natural gaps was generally drier than the anthropogenic gaps (see Humber 2009 for gap-specific data).

Soil temperature at 10 cm depth, as measured during the 2007 growing season, was significantly different among treatments ($F_{2,64} = 44.22$, $P < 0.0001$) but not among disturbance regimes ($F_{1,5} = 0.00$, $P = 0.9702$).

Soil temperatures in T ($13.65 \pm 0.33^\circ\text{C}$) were significantly higher than temperatures in UG ($12.61 \pm 0.33^\circ\text{C}$; $P = 0.0008$) and UE ($10.78 \pm 0.34^\circ\text{C}$, $P < 0.0001$).

Discussion

Invasion of non-native *C. arvense* into boreal forest gaps negatively affects balsam fir regenerative processes at early stages of emergence and survival, and thus may impede natural balsam fir regeneration and the formation of a closed canopy forest independent of future moose densities. However, the success of balsam fir within thistle-invaded boreal forest gaps is dependent upon numerous factors including the life stage in which fir is planted (i.e. as seeds or seedlings), the way in which success is measured (i.e. emergence, survival, growth), the history of forest disturbance, and the distinctive abiotic conditions specific to individual gaps. Older seedlings transplanted into *C. arvense* patches were not disadvantaged over the first year and thus may be a viable management strategy to encourage regeneration in invaded gaps. This knowledge is highly valuable in helping to identify appropriate ecosystem management actions in thistle-invaded boreal forests.

Balsam fir emergence in natural gaps was three times lower in *C. arvense* monocultures than in uninvaded areas of gaps. Because physical removal of either aboveground *C. arvense* biomass or the entire plant did not significantly improve fir emergence, the results suggest that *C. arvense* does not limit fir recruitment through direct competition, but may be the

result of allelopathy or numerous other potential soil properties and microsite conditions (e.g. nutrient availability, water availability, level of compaction). Although fir seedling survival in thistle monocultures was low over the first 2 years, this exotic invader facilitated fir survival over the first growing season such that newly emerged seedlings experienced less than one fifth of the level of browsing, and had over twice the survival rate, than seedlings in uninvaded areas of the gap. This finding is biologically significant since conifer seedlings are at greatest risk of predation during the first 10–16 weeks post-germination (Duchesneau and Morin 1999). During the growing season, the sharp spines on *C. arvense* apparently deterred herbivores and offered protection to neighboring fir from early seedling consumers. Removal of above-ground, or above- and belowground, *C. arvense* biomass was not beneficial as it caused increases in the level of browsing on fir and decreased survival. During the autumn–winter period, *C. arvense* shoots die back and likely offer less protection to fir: browsing incidence on seedlings in all three thistle treatments (T, T_{-A}, and T_{-AB}) increased during this period but damage remained the lowest in the thistle monoculture (T) after 1 year. This facilitative effect was unexpected although similar positive interactions between neighboring plants occurring under conditions of strong consumer pressure have been elsewhere described (Bertness and Callaway 1994; Brooker et al. 2008). A tradeoff may therefore exist for newly emerged seedlings growing amongst *C. arvense*: there may be a net benefit on seedlings at early stages when the seedling is most vulnerable to herbivory, but over time, negative influences on survival could outweigh any benefits. The steady, complete decline in fir survival over 2 years which was observed in the thistle-invaded areas differs from patterns in uninvaded areas of the gap and the forest edge as well as patterns described in other seedling mortality studies (Burns and Honkala 1990; Simard et al. 2003), where initial seedlings losses were the greatest during the first few months but declined only minimally thereafter.

Herbivory by small mammals and slugs was the main cause of newly emerged seedling mortality in uninvaded treatments as well as in the entire *C. arvense* biomass removal treatment; at least 80% of deaths in these treatments could be attributed to either herbivory or desiccation. This early herbivory was likely by non-native species, as only one species (*Deroceras laeve*

Müller) out of ten species of slugs established in GMNP is native (Moss and Hermanutz 2010) and over 90% of small mammal captures in the forests of GMNP have been non-native species (Burzynski et al. 2005). In unaltered thistle monocultures and where thistle shoots were removed, a large proportion of seedlings disappeared without a trace and thus cause of death could not be determined. The seedlings may have been buried and smothered by senescing thistle shoots in autumn, or their mortality increased due to growing under dense cover (Duchesneau and Morin 1999) or allelopathic effects of thistle (Kazinczi et al. 2001; Glinwood et al. 2004; Humber 2009).

Compared with newly emerged seedlings, larger nursery-derived seedlings (aged 15 months at planting) experienced low levels of herbivory (<6%) over the first growing season as they were not browsed by small mammals and slugs and they largely avoided moose herbivory until the autumn–winter period, when moose begin to rely heavily on balsam fir (Des Meules 1962). Despite low levels of herbivory during this growing season, there was some evidence of *C. arvense* protecting these older fir seedlings from herbivory. Over the duration of 1 year, there was no evidence of any negative impact of transplanting nursery-derived balsam fir seedlings into *C. arvense* monocultures. Levels of desiccation were lower, and percent soil moisture contents higher, in thistle-invaded areas of the gap than in native gap vegetation. As moisture is not normally a major limiting factor for *C. arvense* and it commonly grows in very dry soil such as sand dunes and sandy fields (Moore 1975; Nuzzo 1997), this pattern is most likely explained by a favorable moist microclimate created by the dense herbaceous *C. arvense* canopy. The results suggest that older, nursery-derived balsam fir seedlings are capable of growing among dense *C. arvense* monocultures without experiencing significant consequences to survival or growth. Removal of aboveground *C. arvense* shoots by cutting appeared to be slightly disadvantageous to fir seedling success. However, as the success of these transplanted seedlings could be followed for only 1 year, it is not certain whether these benefits will persist over longer time periods. Allelopathic effects of other plant species can cause the failure of tree regeneration (Fisher 1987) therefore it is important to better understand the potential allelopathic interactions between *C. arvense* and native boreal tree species (Humber 2009).

This study revealed that success of balsam fir emergence and establishment differs greatly between disturbance regimes. Transplanted nursery seedlings experienced minimal height growth and increased rates of desiccation-related mortality in natural gaps compared with anthropogenic gaps, yet fir emergence was ubiquitously low throughout all treatments in anthropogenic gaps. Seedbed quality and dominant plant communities in insect-disturbed gaps and harvested gaps in GMNP are known to be very similar (Humber 2009) and seeds were always planted within a consistent seedbed. Further research is needed to investigate if the ubiquitously low emergence within anthropogenic gaps may potentially be the result of higher levels of seed predation in this disturbance regime (Simon et al. 1998; Sullivan et al. 1999), as remnants of predated seeds were occasionally observed within experimental plots.

Management implications

Augmenting thistle-invaded anthropogenic forest gaps of GMNP with balsam fir seeds is not a viable approach to encouraging regeneration due to extremely low emergence. Although emergence was much more successful in natural gaps, sowing is likely still an inefficient management approach because of high rates of loss to early seedling mortality, unpredictable success among gaps, and negative impacts of *C. arvense* on emergence and survival. Frequently-used methods of mechanical thistle control in small infestations (i.e. hand cutting or digging up plants; Trumble and Kok 1982; Donald 1990; Dock Gustavsson 1997) did not improve the success of balsam fir emergence and establishment. Moreover, the sowing protocol involved selectively planting individual seeds at a favorable position slightly under the soil surface, yet large-scale restoration normally involves either broadcast sowing, hydro-seeding, or in very large-scale projects, aerial seeding from low-flying aircraft (Davy 2008). Large-scale sowing would, by necessity, be much less selective than in the present study and previous research indicates that the majority of seedbeds existing within these gaps are unfavorable for conifer recruitment (Humber 2009). For these same reasons, the results suggest that even in the event of significantly decreased moose densities, non-regenerating boreal forest gaps in

GMNP (especially those disturbed by timber harvesting) are not likely to regenerate on their own from seed.

Where barriers to the success of sown seeds exist, restoration processes may be greatly accelerated by planting long-lived perennial species to rapidly reestablish populations of important or keystone species (Davy 2008). The results of this study indicate that planting balsam fir seedlings could be a viable approach to actively promoting native forest regeneration and natural successional processes within gaps, regardless of whether they are invaded by *C. arvense* or not. However, the variability of fir seedling success among gaps suggests that the response of balsam fir to *C. arvense* invasion is complex and highly dependent upon individual gap characteristics. For this reason, any restoration program would greatly benefit from initial small-scale trials in target gaps and a detailed evaluation of the suitability of local environmental conditions. Such an ecosystem approach to management, where ecosystems are the conservation targets rather than individual species (Noss 1996), may be much more effective in this system than targeting control efforts towards *C. arvense* alone since the problem of invasion is rooted in large-scale changes to ecosystem processes and disturbance regimes (Chapin et al. 2000; Myers and Bazely 2003).

Even if single-species control of *C. arvense* is desired, any attempts at achieving and maintaining this control in large or widespread infestations may not be feasible or financially realistic (Mosquin 1997; Rejmánek and Pitcairn 2002; Smith et al. 2006). Aggressive chemical or mechanical weed control in protected natural areas can threaten sensitive non-target native species and stimulate further alien plant invasion (Hobbs and Humphries 1995; Zimdahl 2004; Smith et al. 2006). By shifting the management focus towards re-establishing a native forest canopy, control of non-native, shade-intolerant ruderal plants may also be achieved as they are shaded out over time (Mosquin 1997). *C. arvense* is shade intolerant: growth of its seedlings is considerably reduced in 60–70% of full daylight levels (Bakker 1960; Edwards et al. 2000) and even mature plants experience reduced vigour, vegetative spread, and flower development when growing under taller vegetation or in closed canopy woodlands (Bakker 1960; Trumble and Kok 1982; Donald 1990; Tiley 2010). This

shading principle has been successfully practiced in abandoned farmland and old fields in eastern Canada to form dense closed conifer canopies which shade out alien grasses and forbs over many years (Fisher 1987; Mosquin 1997). This less 'invasive' option to combat large or widespread invasive weed populations holds great potential for the management of sensitive natural areas.

Ultimately however, no restoration program initiated to encourage balsam fir regeneration and canopy formation in Gros Morne National Park will be successful unless moose densities are first lowered. Although there were no negative impacts of *C. arvense* on transplanted fir seedlings over their first year, seedlings in all treatments experienced very high levels of moose herbivory and significant height suppression (see Humber 2009). Allocating valuable resources towards gap regeneration will be futile as long as moose continue to suppress canopy development; similarly, any attempts to control or prevent *C. arvense* invasions will be unsuccessful in the long term if the root cause of the invasion (i.e. overabundant moose) remains. Restoration actions that combine moose reductions with a seedling planting program to restore natural regenerative processes (McLaren et al. 2009) may be the best first step for managers of Gros Morne National Park. In Point Pelee National Park (Canada), culling of overabundant white-tailed deer herds combined with reintroductions of mid-successional trees and shrubs, passive restoration (i.e. encouraging natural colonization), and modifications to topography and hydrology have reduced browsing suppression of Carolinian forests (McLachlan 1997; Buckley et al. 2008). As a result, native herb and shrub communities are gradually increasing and are shading out non-native ruderal plants in some areas (McLachlan 1997; Buckley et al. 2008). Until the overabundant moose population is lowered in Gros Morne National Park, either naturally or through human intervention, these charismatic non-native animals will remain the greatest inhibitors to natural forest regeneration and active restoration alike and will continue to redefine the community compositions of this protected forest landscape.

Acknowledgments Funding for this study was provided by Parks Canada, Memorial University of Newfoundland, and a Canada Graduate Scholarship from the Natural Sciences and

Engineering Research Council of Canada (to JH). Logistical and/or field support was provided by Carson Wentzell, Tom Knight, Kelly Humber, Karen Kennedy, Maria Stapleton, Cecil Oates, Myles Bennett, Shawn Gerrow, Peter Deering, Randy Thompson, Scott Taylor, Michael Burzynski, Michael Henry, Clinton Bennett, and others at Gros Morne National Park. Statistical advice was provided by Keith Lewis and David Schneider. Paul Marino, Tom Knight, Karyn Butler, and Dawn Bazely provided helpful advice during editing stages of manuscript preparation. Thanks to Barry Linehan of the Wooddale Provincial Nursery for donating balsam fir seedlings and to Dale Simpson of the National Tree Seed Centre for providing seeds for this project.

References

- Bakker D (1960) A comparative life-history study of *Cirsium arvense* (L.) Scop., *Tussilago farfara* L., the most troublesome weeds in the newly reclaimed polders of the former Zuiderzee. In: Harper JL (ed) The biology of weeds. Blackwell Scientific Publications, Oxford, pp 205–222
- Bendall GM (1975) The allelopathic activity of Californian thistle (*Cirsium arvense* (L.) Scop.) in Tasmania. Weed Res 15(22):77–81
- Bertness M, Callaway RM (1994) Positive interactions in communities. Trends Ecol Evol 9:191–193
- Bouchard A, Hay SG, Bergeron Y, Leduc A (1991) The vascular flora of Gros Morne National Park, Newfoundland: a habitat classification approach based on floristic, biogeographical, and life-form data. In: Nimis PL, Covello TJ (eds) Quantitative approaches to phytogeography. Kluwer Academic, The Netherlands, pp 123–157
- Brooker RW, Maestre FT, Callaway RM, Lortie CL, Cavieres LA, Kunstler G, Liancourt P, Tielbörger K, Travis JM, Anthelme F, Armas C, Coll L, Corcket E, Delzon S, Forey E, Kikvidze Z, Olofsson J, Pugnaire F, Quiroz CL, Saccone P, Schiffrers K, Seifan M, Touzard B, Michalet R (2008) Facilitation in plant communities: the past, the present, and the future. J Ecol 96:18–34
- Brothers TS, Spingarn A (1992) Forest fragmentation and alien plant invasion of central Indiana old-growth forests. Conserv Biol 6:91–100
- Buckley P, Ito S, McLachlan S (2008) Temperate woodlands. In: Perrow MR, Davy AJ (eds) Handbook of ecological restoration, volume 2: restoration in practice. Cambridge University Press, New York, pp 503–538
- Burke MJW, Grime JP (1996) An experimental study of plant community invasibility. Ecology 77:776–790
- Burns RM, Honkala BH (1990) Silvics of North America: 1. Conifers; 2. Hardwoods. Agriculture handbook 654, US Department of Agriculture, Forest Service, Washington, DC
- Burzynski M, Knight T, Gerrow S, Hoffman J, Thompson R, Deering P, Major D, Taylor S, Wentzell C, Simpson A, Burdett W (2005) State of the park report: an assessment of ecological integrity. Gros Morne National Park of Canada

- Chapin FS, Zavaleta ES, Eviner VT, Naylor RL, Vitousek PM, Reynolds HL, Hooper DU, Lavorel S, Sala OE, Hobbie SE, Mack MC, Díaz S (2000) Consequences of changing biodiversity. *Nature* 405:234–242
- Clayton JS, Erlich WA, Cann DB, Day JH, Marshall IB (1977) Soil report. Agriculture Canada, Ottawa
- Coomes DA, Allen RB, Forsyth DM, Lee WG (2003) Factors preventing the recovery of New Zealand forests following control of invasive deer. *Conserv Biol* 17:450–459
- Cowie ID, Werner PA (1993) Alien plant species invasive in Kakadu National Park, tropical northern Australia. *Biol Conserv* 63:127–135
- Davy AJ (2008) Establishment and manipulation of plant populations and communities in terrestrial systems. In: Perrow MR, Davy AJ (eds) *Handbook of ecological restoration, volume 1: principles of restoration*. Cambridge University Press, New York, pp 223–241
- Des Meules P (1962) Intensive study of an early spring habitat of moose (*Alces alces americana* Cl.) in Laurentide Park. Proceedings of the conference on northeast wildlife, Québec
- Dock Gustavsson AM (1997) Growth and regenerative capacity of plants of *Cirsium arvense*. *Weed Res* 37:229–236
- Donald WW (1990) Management and control of Canada thistle (*Cirsium arvense*). *Rev Weed Sci* 5:193–249
- Donald WW (1994) The biology of Canada thistle (*Cirsium arvense*). *Rev Weed Sci* 6:77–101
- Duchesneau R, Morin H (1999) Early seedling demography in balsam fir seedling banks. *Can J For Res* 29:1502–1509
- Edwards GR, Bourdôt GW, Crawley MJ (2000) Influence of herbivory, competition and soil fertility on the abundance of *Cirsium arvense* in acid grassland. *J Appl Ecol* 37:321–334
- Fai AHT, Cornelius PL (1996) Approximate F-tests of multiple degree of freedom hypotheses in generalized least squares analyses of unbalanced splitplot experiments. *J Stat Comput Simul* 54:363–378
- Fisher RF (1987) Allelopathy: a potential cause of forest regeneration failure. In: Waller GR (ed) *Allelochemicals: role in agriculture and forestry*. American chemical society symposium series 330, Washington, DC, pp 176–184
- Forbes G (2006) Assessment of information needs regarding moose management in Gros Morne and Terra Nova National Parks, Newfoundland. New Brunswick Cooperative Fish and Wildlife Research Unit, University of New Brunswick, Fredericton, NB
- Glinwood R, Ninkovic V, Pettersson J, Ahmed E (2004) Barley exposed to aerial allelopathy from thistles (*Cirsium* spp.) becomes less acceptable to aphids. *Ecol Entomol* 29:188–195
- Gosse J (2006) Moose-vegetation issues in Terra Nova and Gros Morne National Parks; options for active management. Unpublished report, Parks Canada, Glovertown, NL
- Gosse J, Hermanutz L, McLaren B, Deering P, Knight T (2011) Degradation of boreal forests by non-native herbivores in Newfoundland's National Parks: recommendations for ecosystem restoration. *Nat Areas J* (in press)
- Greene DF, Zasada JC, Sirois L, Kneeshaw D, Morin H, Charron I, Simard M-J (1999) A review of the regeneration dynamics of North American boreal forest tree species. *Can J For Res* 29:824–839
- Haber E (1997) Impact of invasive plants on species and habitats at risk in Canada. Canadian Wildlife Service, Environment Canada, Ottawa
- Hall JP, Richardson J (1973) Natural regeneration after disturbance in the forests of western Newfoundland. Newfoundland Forest Research Centre Information report N-X-92
- Hartley SE, Jones CG (1997) Plant chemistry and herbivory: or why the world is green. In: Crawley MJ (ed) *Plant ecology*, 2nd edn. Blackwell Science, Oxford, pp 284–324
- Hendrickson C, Bell T, Butler K, Hermanutz L (2005) Disturbance-enabled invasion of *Tussilago farfara* (L.) in Gros Morne National Park, Newfoundland: management implications. *Nat Areas J* 25:263–274
- Hobbs RJ, Humphries SE (1995) An integrated approach to the ecology and management of plant invasions. *Conserv Biol* 9:761–770
- Hobbs RJ, Norton DA (1996) Towards a conceptual framework for restoration ecology. *Restor Ecol* 4:93–110
- Horton NJ, Lipsitz SR (1999) Review of software to fit generalized estimating equation regression models. *Am Stat* 53:160–169
- Humber JM (2009) Non-native plant invasion of boreal forest gaps: implications for stand regeneration in a protected area shaped by hyperabundant herbivores. MSc thesis, Memorial University of Newfoundland
- Kazinczi G, Beres I, Narwal SS (2001) Allelopathic plants. 1. Canada thistle (*Cirsium arvense* (L.) Scop.). *Allelopathy J* 8:29–40
- Levine JM, Vilà M, D'Antonio CM, Dukes JS, Grigulis K, Lavorel S (2003) Mechanisms underlying the impacts of exotic plant invasions. *Proc R Soc Lond B* 270(1517):775–781
- Littell RC, Stroup WW, Freund RJ (2002) SAS for linear models, 4th edn. SAS Institute Inc., Cary
- Macdonald IAW, Kruger FJ, Ferrar AA (1986) The ecology and management of biological invasion in Southern Africa. In: Proceedings of the national synthesis symposium on the ecology of biological invasions. Oxford University Press, Cape Town, pp 251–276
- Mataczyk JA, Willis AJ, Vranjic JA, Ash JE (2002) Herbicides, weeds, and endangered species: management of bitou bush (*Chrysanthemoides monilifera rotundata*) with glyphosate and impacts on the endangered shrub, *Pimelea spicata*. *Biol Conserv* 108:133–141
- McCullagh P, Nelder JA (1999) Generalized linear models, 2nd edn. Chapman and Hall/CRC, Boca Raton
- McLachlan SM (1997) Multiple-scale approaches to the restoration of deciduous forest in southwestern Ontario, Canada. PhD dissertation, York University
- McLaren B, Tom-Dery D (2007) Ungulate control in protected areas: a review of options. Parks Canada unpublished report
- McLaren B, McCarthy C, Mahoney S (2000) Extreme moose demographics in Gros Morne National Park, Newfoundland. *Alces* 36:217–232
- McLaren BE, Roberts BA, Djan-Chékar N, Lewis KP (2004) Effects of overabundant moose on the Newfoundland landscape. *Alces* 40:45–59
- McLaren BE, Hermanutz L, Gosse J, Collet B, Kasimos C (2009) Broadleaf competition interferes with balsam fir

- regeneration following experimental removal of moose. *For Ecol Manag* 257:1395–1404
- Messing RH, Wright MG (2006) Biological control of invasive species: solution or pollution? *Front Ecol Environ* 4:132–140
- Moore RJ (1975) The biology of Canadian weeds. 13. *Cirsium arvense* (L.) Scop. *Can J Plant Sci* 55:1033–1048
- Morin H, Laprise D (1997) Seedling bank dynamics in boreal balsam fir forests. *Can J For Res* 27:1442–1451
- Mosquin T (1997) Management guidelines for invasive alien species in Canada's National Parks. National Parks Branch, Parks Canada, Ottawa
- Moss M, Hermanutz L (2010) Monitoring the small and slimy—protected areas should be monitoring native and non-native slugs (Mollusca: Gastropoda). *Nat Areas J* 30(3):322–327
- Myers JH, Bazely DR (2003) Ecology and control of introduced plants: evaluating and responding to invasive plants. Ecology, biodiversity and conservation series, Cambridge University Press, Cambridge
- Noss RF (1996) Ecosystems as conservation targets. *Trends Ecol Evol* 11:351
- NRC (National Research Council) (2000) The future role of pesticides in US agriculture. National Academy Press, Washington
- Nuzzo V (1997) Element stewardship abstract: *Cirsium arvense*. The nature conservancy wildland invasive species program <http://tncinvasives.ucdavis.edu/esadocs/documnts/cirsarv.pdf>. Accessed 9 Feb 2009
- Parks Canada Agency (2000) Unimpaired for future generations? Protecting ecological integrity with Canada's National Parks, vol. II, setting a new direction for Canada's National Parks. Report of the panel on the ecological integrity of Canada's National Parks, Ottawa
- Pearson DE, Callaway RM (2003) Indirect effects of host-specific biological control agents. *Trends Ecol Evol* 18:456–461
- Rejmánek M, Pitcairn MJ (2002) When is eradication of exotic pest plants a realistic goal? In: Veitch CR, Clout MN (eds) Turning the tide: the eradication of invasive species. Occasional paper of the IUCN species survival commission no. 27, pp 249–253
- Rose MD (2002) Are boreal ecosystems susceptible to invasion by alien plants? A case study of Gros Morne National Park. MSc thesis, Memorial University of Newfoundland
- Rose M, Hermanutz L (2004) Are boreal ecosystems susceptible to alien plant invasion? Evidence from protected areas. *Oecologia* 139:467–477
- Siemens DH, Garner SH, Mitchell-Olds T, Callaway RM (2002) Cost of defense in the context of plant competition: *Brassica rapa* may grow and defend. *Ecology* 83:505–517
- Simard M, Bergeron Y, Sirois L (2003) Substrate and litterfall effects on conifer seedling survivorship in southern boreal stands of Canada. *Can J For Res* 33:672–681
- Simon NPP, Schwab FE, Baggs EM, GIMcT Cowan (1998) Distribution of small mammals among successional and mature forest types in western Labrador. *Can Field Nat* 112(3):441–445
- Skinner K, Smith L, Rice P (2000) Using noxious weed lists to prioritize targets for developing weed management strategies. *Weed Sci* 48:640–644
- Smith RG, Maxwell BD, Menalled FD, Rew LJ (2006) Lessons from agriculture may improve the management of invasive plants in wildland systems. *Front Ecol Environ* 4(8):428–434
- Sokal RR, Rohlf JF (1995) Biometry: the principles and practices of statistics in biological research. W.H. Freeman and Company, New York
- Stowe LG (1979) Allelopathy and its influence on the distribution of plants in an Illinois old field. *J Ecol* 67:1065–1085
- Sullivan TP, Lautenschlager RA, Wagner RG (1999) Clearcutting and burning of northern spruce-fir forests: implications for small mammal communities. *J Appl Ecol* 36:327–344
- Thompson RG (2007) Gros Morne National Park 2007 aerial moose survey. Unpublished report. Parks Canada, Rocky Harbour, NL
- Tiley GED (2010) Biological flora of the British Isles: *Cirsium arvense* (L.) Scop. *J Ecol* 98:938–983
- Trumble JT, Kok LT (1982) Integrated pest management techniques in thistle suppression in pastures of North America. *Weed Res* 22:345–359
- Vitousek PM, D'Antonio CM, Loope LL, Rejmanek M, Westbrooks R (1997) Introduced species: a significant component of human-caused global change. *N Z J Ecol* 21:1–16
- Wilson JR (1981) Effect of Canada thistle (*Cirsium arvense*) residue on growth of some crops. *Weed Sci* 29:159–164
- Zavaleta ES, Hobbs RJ, Mooney HA (2001) Viewing invasive species removal in a whole-ecosystem context. *Trends Ecol Evol* 16(8):454–459
- Zimdahl RL (2004) Weed-crop competition: a review, 2nd edn. Blackwell Publishing, Ames